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bhavan@LAPTOP-FGDPB110:~$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
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warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

bhavan@LAPTOP-FGDPB110:~$ nano process.c
bhavan@LAPTOP-FGDPB110:~$ gcc process.c -o process
bhavan@LAPTOP-FGDPB110:~$ ./process
Hello, World!
bhavan@LAPTOP-FGDPB110:~$ ls
process  process.c  process.c.save  process.c.save.1
bhavan@LAPTOP-FGDPB110:~$ pwd
/home/bhavan
bhavan@LAPTOP-FGDPB110:~$ date
Wed Aug  5 09:22:21 UTC 2026
bhavan@LAPTOP-FGDPB110:~$ cal
Command 'cal' not found, but can be installed with:
sudo apt install ncal
bhavan@LAPTOP-FGDPB110:~$ whoami
bhavan
bhavan@LAPTOP-FGDPB110:~$ hostname
LAPTOP-FGDPB110
bhavan@LAPTOP-FGDPB110:~$ uname
Linux
bhavan@LAPTOP-FGDPB110:~$ uname -a
Linux LAPTOP-FGDPB110 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
bhavan@LAPTOP-FGDPB110:~$ ^C
bhavan@LAPTOP-FGDPB110:~$ ps
  PID TTY          TIME CMD
   320 pts/0    00:00:00 bash
  4223 pts/0    00:00:00 ps
bhavan@LAPTOP-FGDPB110:~$ !

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top - 09:24:47 up 19 min, 1 user, load average: 0.06, 0.06, 0.07
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.7 sy,  0.0 ni, 98.6 id,  0.7 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7754.6 total,  7133.6 free,   540.4 used,   240.3 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7214.3 avail Mem

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PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	22132	13508	9732	S	0.0	0.2	0:04.73	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.05	init-systemd(Ub
6	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00	init
58	root	19	-1	50440	16300	15196	S	0.0	0.2	0:00.56	systemd-journal
107	root	20	0	25284	6608	5172	S	0.0	0.1	0:00.75	systemd-udev
120	systemd+	20	0	21344	13128	11008	S	0.0	0.2	0:00.26	systemd-resolve
121	systemd+	20	0	91036	7992	7024	S	0.0	0.1	0:00.17	systemd-timesyn
181	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.01	cron
182	message+	20	0	9836	5620	4680	S	0.0	0.1	0:00.66	dbus-daemon
193	root	20	0	17976	8884	7844	S	0.0	0.1	0:00.38	systemd-logind
215	root	20	0	3124	1976	1836	S	0.0	0.0	0:00.01	agetty
227	root	20	0	107032	23220	13768	S	0.0	0.3	0:00.28	unattended-upgr
318	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
319	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.17	Relay(320)
320	teja	20	0	6080	5392	3672	S	0.0	0.1	0:00.15	bash
321	root	20	0	6700	4636	3856	S	0.0	0.1	0:00.01	login
368	teja	20	0	20320	11500	9380	S	0.0	0.1	0:00.27	systemd
369	teja	20	0	21160	3584	1860	S	0.0	0.0	0:00.00	(sd-pam)
398	teja	20	0	6080	5360	3692	S	0.0	0.1	0:00.04	bash
682	polkitd	20	0	308172	8012	7104	S	0.0	0.1	0:00.18	polkitd
3640	syslog	20	0	222516	5508	4588	S	0.0	0.1	0:00.06	rsyslogd
4230	teja	20	0	9308	5444	3256	R	0.0	0.1	0:00.01	top

```
2 clear
3 sudo apt update
4 sudo apt install build-essential installing:
5 gcc --version
6 gcc process . c -o process
7 ./process
8 clear
9 mkdir OSSP
10 cd OSSP
11 git init
12 mkdir src include obj bin docs tests scripts assets
13 touch .gitignore README.md LICENSE Makefile
14 touch src/main.c
15 touch src/shell.c
16 touch src/parser.c
17 touch src/executor.c
18 touch src/builtin.c
19 touch include/shell.h
20 touch include/parser.h
21 touch include/executor.h
22 touch include/builtin.h
23 touch docs/design.md
24 touch docs/report.pdf
25 touch tests/test_parser.c
26 touch tests/test_executor.c
27 touch scripts/run.sh
28 touch assets/architecture.png
29 tree
30 sudo apt install tree
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40 clear
41 pwd
42 clear
43 clearhistory
44 clear
45 history
46 whoami
47 date
48 cal
49 hostname
50 uname -a
51 exit
52 sudo apt update
53 sudo apt upgrade
54 sudo apt install gcc
55 gcc --version
56 nano process.c
57 gcc process.c -o process
58 ./process
59 ls
60 pwd
61 date
62 cal
63 whoami
64 hostname
65 uname
66 uname -a
67
```

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52 sudo apt update
53 sudo apt upgrade
54 sudo apt install gcc
55 gcc --version
56 nano process.c
57 gcc process.c -o process
58 ./process
59 ls
60 pwd
61 date
62 cal
63 whoami
64 hostname
65 uname
66 uname -a
67 ps
68 top
69 history
bhavan@APT0P-FGDPB110:~$ clear
```

1. **sudo apt update**

Updates the package information from the configured repositories.

2. **sudo apt upgrade**

Upgrades the installed packages to their latest available versions.

3. **sudo apt install gcc**

Installs the **GCC (GNU Compiler Collection)** compiler required to compile C programs.

4. **gcc --version**

Displays the installed GCC compiler version.

5. **nano process.c**

Opens the process.c file using the **Nano text editor**. It can be used to create or edit a C prog